CSE 499A Progress Report

Multimodal AI for Cause-of-Death Estimation from Postmortem Data

Course	CSE 499A
Group	2
Supervisor	Dr. Sumaiya Tabassum Nimi [STI]
Department	Electrical and Computer Engineering
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1. Project Overview

Our project is about estimating the cause of death from postmortem data using machine learning. We use the New Mexico Decedent Image Database (NMDID), which has both decedent metadata (age, sex, medical history, medications, and so on) and postmortem CT (PMCT) images. We chose NMDID because it is one of the few large collections where these two kinds of data are linked for the same person.

The long-term idea is to combine the metadata and the PMCT images in one model (multimodal fusion). Metadata can give background information about a person, while the images can show injuries or other findings. We think the two together may predict better than either one alone, but we have not tested this yet. So far, our work is on the metadata side only.

Each of the 860 decedents is placed in one of five cause-of-death categories:

- Blunt trauma
- Firearm
- Stab
- Cardiac
- Intoxication

2. Previous Work Summary

The 28 October report covered the first metadata-only experiments. In short, we:

- Prepared the metadata for 860 decedents.
- Organized 70 numerical features into six groups.
- Implemented TabPFN 3.5, Logistic Regression, and Random Forest.
- Compared two feature sets.
- Completed a class-wise performance analysis.
- Completed the age distribution analysis and age-ablation experiments.

The best repeated cross-validation result from that report was **Random Forest with 70 features, Macro-F1 = 0.3399 ± 0.0300**. Macro-F1 is the average F1 score over all classes, so small classes count as much as large ones. This number is a baseline, and it was measured differently from the validation results in Section 3.D.

3. New Work Completed

This section covers the work done since the last report, in the order we did it. All of it uses only the training and validation partitions. The independent test set has not been evaluated.

A. Label Mapping and Data Splitting

Besides the five original classes, we defined two extra tasks by grouping labels. The broad three-class task merges the three trauma types into one group. The trauma subtype task only separates the three kinds of trauma. The label mapping is saved in the repository as a CSV file.

Table 1. Classification tasks and class counts.

Task	Classes and counts (860 records)
Five-class	Blunt trauma 149, Firearm 64, Stab 79, Cardiac 202, Intoxication 366
Broad three-class	Trauma 292, Cardiac 202, Intoxication 366
Trauma subtype	Blunt trauma 149, Firearm 64, Stab 79 (trauma records only)

We then fixed the data split so that every experiment uses the same records. The split is stratified, so each part keeps about the same class percentages.

Table 2. Fixed data split.

Dataset	Records
Training	602
Validation	129
Testing	129
Total	860

The 129-record independent test set has **not been evaluated**. We are keeping it untouched until the protocol is frozen.

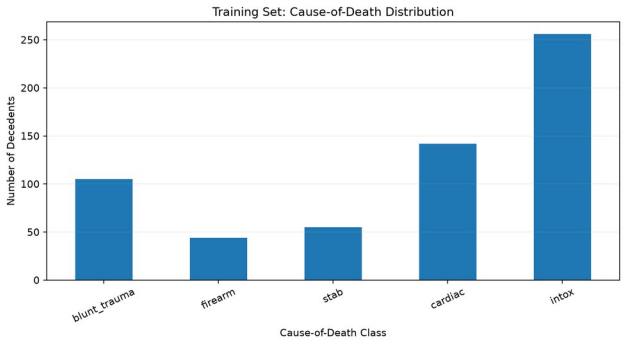


Figure 1. Class distribution in the training set (602 records). Intoxication is the largest class and firearm the smallest.

B. Preliminary Leakage Audit

Feature leakage means a model uses information that would not really be available when making the prediction, or that is only recorded because of the cause of death. A model with leaked features can look good on paper and still be useless in practice, so we need to check each feature's source and timing.

We did a first classification of the 70 features:

- **2 candidate allowed features.**
- **68 gray features** that need their source and recording time verified.
- **2 forbidden columns** outside the feature set: the decedent ID and the target label.

This is a **preliminary classification only**. It is not a fully verified leakage audit, and most features still need to be checked against the data dictionary.

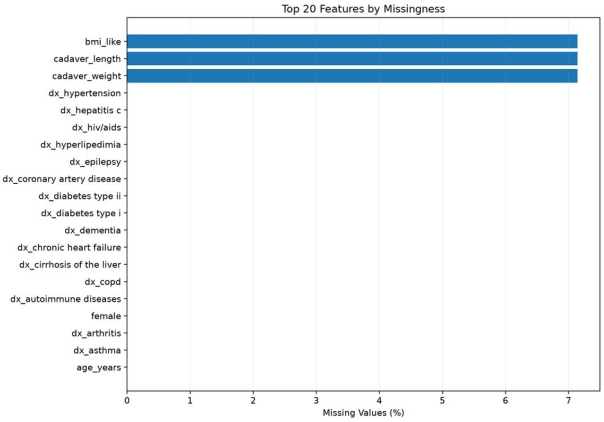


Figure 2. Top 20 features by missing values. Only bmi_like, cadaver_length, and cadaver_weight have missing values (about 7%); the rest are almost complete.

C. Exploratory Data Analysis and Statistical Tests

On the 602 training records we looked at feature distributions and missing values. To see which features are linked to each class, we ran Fisher's exact test for binary features and the Mann–Whitney U test for continuous features, each class against the rest. Because this makes many tests, we used Benjamini–Hochberg FDR correction (FDR means false discovery rate, which controls how many of the "significant" results are expected to be false alarms). We also ran age/sex-adjusted logistic regression and made odds-ratio heatmaps and a correlation plot.

Table 3. Summary of the statistical tests (350 = 70 features × 5 classes).

Analysis	Result
Training records	602
Features	70
Unadjusted tests	350
Unadjusted FDR-significant tests	34
Adjusted models attempted	350
Adjusted models estimated	244
Adjusted FDR-significant tests	25

Of the 350 adjusted models, 106 could not be estimated reliably (flagged as unstable or possible separation), so only 244 were used. A statistical association does not show causation, and it does not show that a feature is safe to use.

D. Leakage-Scenario Model Comparison

To see how much the result depends on the uncertain features, we grouped the features into nine scenarios, from S0 (demographics only, 2 features) up to S8 (all 70 features). The other scenarios are basic features, diagnoses, medical history, substance use, lifestyle, medications, and all history features (S7, 67 features). We tested three models (Logistic Regression, Random Forest, LightGBM) on three tasks (five-class, broad three-class, trauma subtype). That gives 9 × 3 × 3 = **81 validation experiments**, all completed.

Table 4. Best validation Macro-F1 for each task.

Task	Model	Scenario	Macro-F1
Five-class	Random Forest	S7_all_history	0.3698
Broad three-class	LightGBM	S7_all_history	0.5753
Trauma subtype	LightGBM	S6_medications	0.4684

These are **validation results used to compare models**. They are not independent test results, and they should not be compared directly with the repeated cross-validation number from the last report (0.3399). The best configuration was picked using the same validation set, so these numbers are probably a bit optimistic. The differences between scenarios are small for the five-class task, and the trauma subtype task has only 44 validation records.

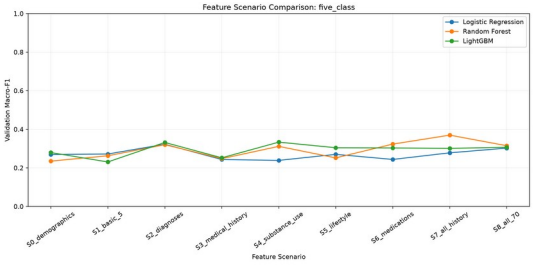


Figure 3. Validation Macro-F1 across the nine scenarios, five-class task.

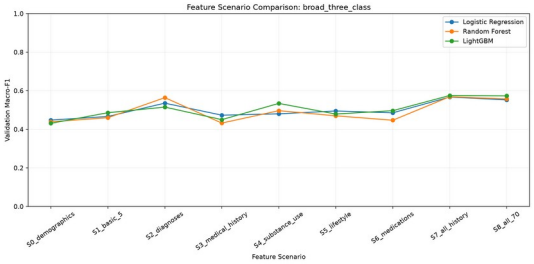


Figure 4. Validation Macro-F1 across the nine scenarios, broad three-class task.

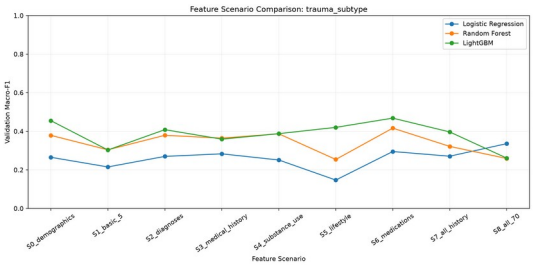


Figure 5. Validation Macro-F1 across the nine scenarios, trauma subtype task.

E. SHAP Explainability

SHAP is a method that shows how much each feature pushed a model's prediction up or down. We used TreeSHAP on the three best models from Table 4. In the table below, "unverified history-related importance" is the share of the total SHAP importance that comes from the diagnosis, history, substance, lifestyle, and medication features, whose source and timing are not yet verified.

Table 5. Global SHAP findings for the selected models.

Task	Important features	Unverified history-related importance
Five-class	med_na, age_years, n_meds	85.9%
Broad three-class	age_years, med_na, su_na	74.4%
Trauma subtype	age_years, med_na, n_meds	23.5%

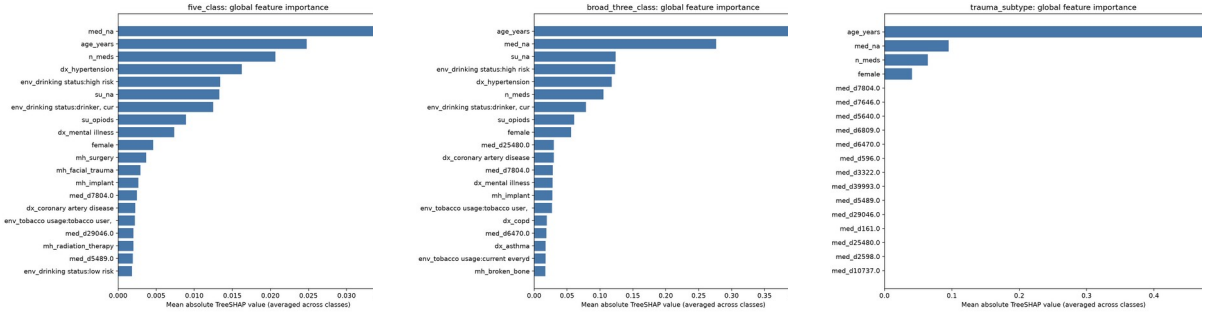


Figure 6. Global SHAP importance (mean absolute value). Left to right: five-class, broad three-class, trauma subtype.

Here, med_na, su_na, and n_meds look like missingness indicators or counts (for example, whether medication data are missing, and how many medications are listed). They rank very high, which means the models partly rely on how the data were recorded, not on real medical information. We still need to check these against the data dictionary. SHAP shows what the model relies on. It does not show causes, and it does not show that a feature is safe.

F. Probability Calibration

A model is well calibrated if its predicted probabilities match real outcomes, for example, cases given 70% should be correct about 70% of the time. We applied temperature scaling, which divides the model's output scores by one number (the temperature) so the probabilities become less or more confident. The temperature was fitted on a separate part of the training set, not on the validation set. We measured Brier score and log loss (lower is better for both) and ECE (expected calibration error, the gap between confidence and accuracy, also lower is better).

Table 6. Validation metrics before → after temperature scaling.

Task	Temp.	Brier	Log loss	ECE
Five-class	0.960	0.679 → 0.678	1.344 → 1.341	0.119 → 0.102
Broad three-class	1.566	0.510 → 0.507	0.869 → 0.854	0.057 → 0.073
Trauma subtype	7.684	0.682 → 0.655	1.129 → 1.081	0.173 → 0.070

Brier score and log loss improved for all three tasks. ECE improved for the five-class and trauma subtype tasks, but it got worse for the broad three-class task, so calibration did not improve on every measure. The changes for the five-class task are very small. The trauma subtype temperature (7.68) was estimated from only 41 records, so it is not reliable.

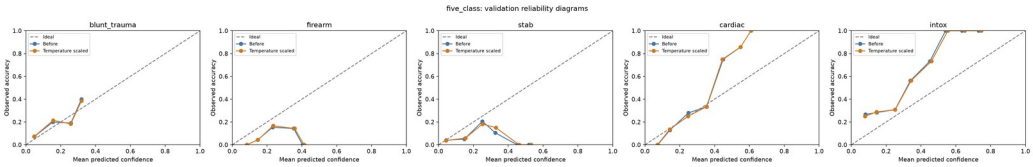


Figure 7. Reliability diagrams on validation, five-class task (one curve per class, before and after temperature scaling). The dashed line is perfect calibration.

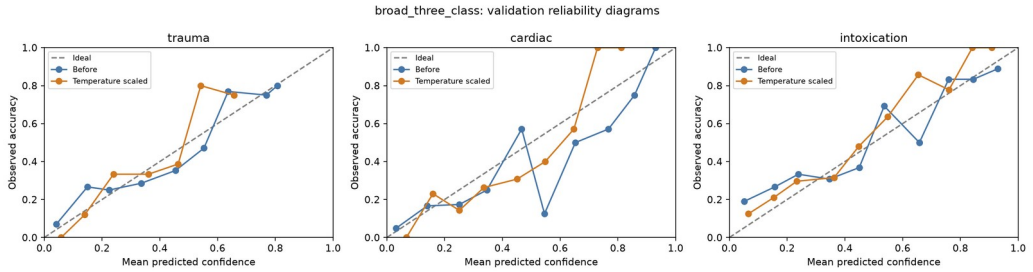


Figure 8. Reliability diagrams on validation, broad three-class task (one curve per class, before and after temperature scaling). The dashed line is perfect calibration.

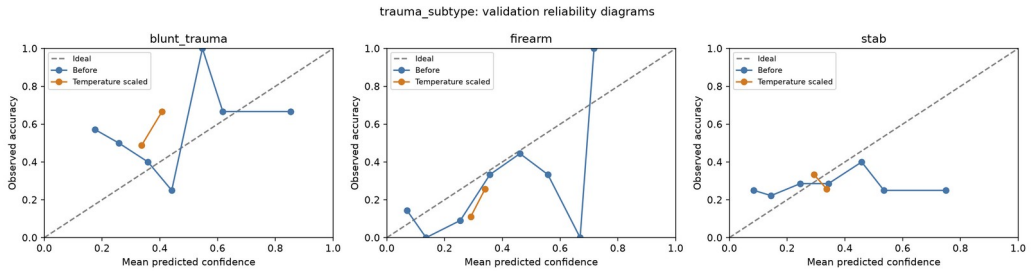


Figure 9. Reliability diagrams on validation, trauma subtype task (one curve per class, before and after temperature scaling). The dashed line is perfect calibration.

G. Conformal Prediction

Conformal prediction lets a model show uncertainty. Instead of always giving one answer, it returns a set of possible classes. For a 90% target, the set is supposed to contain the true class about 90% of the time. We used split conformal prediction, with a separate part of the training set to set the threshold. The table shows how often the true class was inside the set on the validation records (coverage).

Table 7. Validation coverage of the prediction sets.

Task	90% target	80% target
Five-class	85.3%	80.6%
Broad three-class	91.5%	79.8%
Trauma subtype	95.5%	90.9%

The sets were often multi-class. The share of records with a single-class set (singleton) was 0–10.9% at the 90% target and 1.6–34.9% at the 80% target. The models often cannot pick one class at these confidence levels, and the five-class model almost never does.

These coverage numbers are descriptive only, because the same validation set was already used to choose the models. Conformal coverage is marginal under an exchangeability assumption. It is not guaranteed for every class. For example, the five-class stab coverage was 75% at the 90% target, below the target. The trauma subtype calibration set has only 9 firearm cases, so those results are noisy.

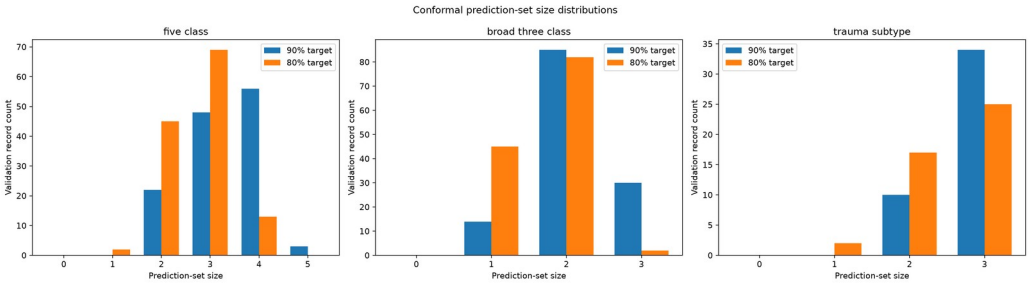


Figure 10. Number of validation records for each prediction-set size, at 90% and 80% targets.

4. Existing Project Pipeline

The figure below shows the full planned pipeline. Solid blue boxes are implemented so far (all exploratory). Dashed boxes have not been started. **PMCT preprocessing, Merlin, and multimodal fusion are not implemented yet.**

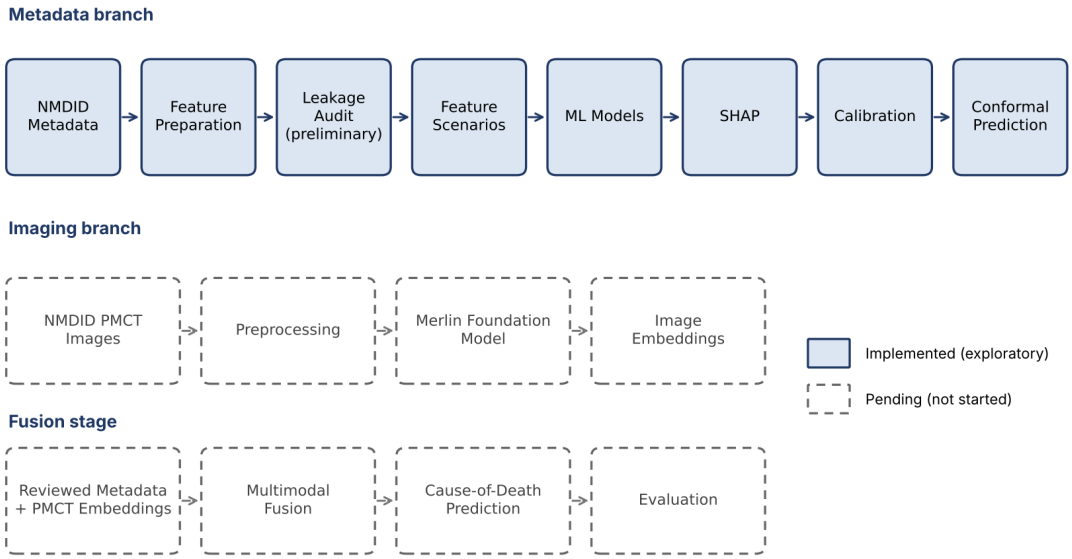


Figure 11. Project pipeline: metadata branch (implemented), imaging branch and fusion stage (pending).

5. Project Structure

Our repository is organized like this (based on the files we have in the repository):

```
project/
├── run_reliability_stage.py
├── requirements.txt
├── docs/
│   └── reliability_analysis.md
├── reports/
│   └── figures/
└── protocol/
    ├── label_map.csv
    ├── split_summary.csv
    ├── eda/
    ├── leakage_scenarios/
    ├── shap/
    ├── calibration/
    ├── conformal/
    └── supervisor_demo/
```

Table 8. Repository folders.

Folder / file	Purpose
run_reliability_stage.py	Runs the SHAP, calibration, conformal, and summary stages in order.
docs/	Written documentation of the analysis.
reports/figures/	Figures from the earlier report (model comparison, age analysis, confusion matrices).
protocol/label_map.csv, split_summary.csv	Label mapping and the fixed train, validation, and test split counts.
protocol/eda/	Statistical test tables and EDA figures.
protocol/leakage_scenarios/	Results and figures for the 81 scenario experiments.
protocol/shap/	SHAP report and plots.
protocol/calibration/	Calibration metrics and reliability diagrams.
protocol/conformal/	Prediction-set metrics, method note, and figures.
protocol/supervisor_demo/	Short summary of the SHAP, calibration, and conformal stage.

The raw NMDID data and the local decedent ID assignments are kept outside the public repository outputs. The saved results only contain aggregate numbers, with no person-level IDs, probabilities, or prediction sets.

6. Current Project Status

Table 9. Project status.

Work	Status
Metadata preparation	Completed
Metadata-only model comparison	Completed
Age analysis and ablation	Completed
Label mapping and fixed splits	Completed
EDA and FDR analysis	Completed
Leakage-scenario experiments	Completed
SHAP	Completed, exploratory
Calibration	Completed, exploratory
Conformal prediction	Completed, exploratory
Feature provenance verification	Pending
Independent test evaluation	Pending
PMCT preprocessing	Pending
Merlin embeddings	Pending
Multimodal fusion	Pending

7. Limitations

- **Class imbalance.** Intoxication is 42.5% of the records and firearm only 7.4%, so small classes are hard to learn and the scores are noisy.
- **Unverified features.** The source and recording time of the history, substance, lifestyle, and medication features have not been checked.
- **Missingness indicators in SHAP.** Features like med_na and su_na rank very high, so the models may use how the data were recorded rather than real information.
- **Small calibration subsets.** The trauma subtype calibration and conformal subsets have only 41 records each, with 9 firearm cases.
- **Validation reuse.** The same validation set was used to compare 81 configurations, so the validation numbers are descriptive and likely optimistic.
- **Untouched test set.** We have no independent test result yet.

- **No imaging work yet.** PMCT preprocessing, Merlin embeddings, and multimodal fusion have not been implemented.

8. Future Work

1. Verify all feature definitions, sources, and recording times.
2. Finalize the allowed, gray, and forbidden feature list.
3. Freeze the experimental protocol.
4. Run the prespecified independent test evaluation once.
5. Obtain approved PMCT images.
6. Test CT preprocessing on a small sample.
7. Extract image embeddings using Merlin.
8. Compare metadata-only, PMCT-only, and multimodal models.
9. Check whether combining images and metadata improves prediction.

9. Conclusion

Since the last report, we fixed the label mapping and data splits, ran EDA with FDR-corrected tests, and compared 81 model and feature-scenario combinations. We also added SHAP, calibration, and conformal prediction analyses. This finishes a large part of the metadata-only stage.

The results are still exploratory. The scores are modest, many features are unverified, and the models rely on some features that may reflect data recording. Our next step is to verify the features and run the single test evaluation. After that, we can move on to PMCT preprocessing, Merlin embeddings, and multimodal fusion, none of which is done yet.